

Session 1 — Probability I

Solution sheet

Complete worked solutions, with reminders of the key ideas. Numerical answers are boxed. Encourage students to reproduce each calculation rather than just read it.

Exercise 1 Two fair dice.

Solution. There are $6 \times 6 = 36$ equally likely outcomes.

(a) Sum = 7: the pairs (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) — six of them. $\mathbb{P} = \frac{6}{36} = \boxed{1/6}$.

(b) Sum = 11: only (5, 6), (6, 5), so $\frac{2}{36}$. These are disjoint from “sum = 7”, so add: $\frac{6}{36} + \frac{2}{36} = \frac{8}{36} = \boxed{2/9}$.

(c) “At least one 6” via the complement: $\mathbb{P}(\text{no 6}) = \left(\frac{5}{6}\right)^2 = \frac{25}{36}$, so $\mathbb{P}(\text{at least one 6}) = 1 - \frac{25}{36} = \boxed{11/36 \approx 0.306}$.

Reminder

For “at least one” it is almost always easier to compute $1 - \mathbb{P}(\text{none})$. Disjoint events (they cannot occur together) let you simply *add* probabilities.

Exercise 2 Target on each of 5 frames, $p = 0.15$, independent.

Solution. $\mathbb{P}(\text{no target}) = (1 - 0.15)^5 = 0.85^5 = 0.4437$, so

$$\mathbb{P}(\text{at least one}) = 1 - 0.85^5 = \boxed{0.5563} \quad (\approx 55.6\%).$$

Exercise 3 120 bilingual, 90 musicians, 50 both (out of 200).

Solution. Work with proportions: $\mathbb{P}(B) = 0.60$, $\mathbb{P}(M) = 0.45$, $\mathbb{P}(B \cap M) = 0.25$.

(a) $\mathbb{P}(B \cup M) = 0.60 + 0.45 - 0.25 = \boxed{0.80}$.

(b) $\mathbb{P}(\text{neither}) = 1 - 0.80 = \boxed{0.20}$.

(c) $\mathbb{P}(\text{exactly one}) = \mathbb{P}(B \cup M) - \mathbb{P}(B \cap M) = 0.80 - 0.25 = \boxed{0.55}$.

Reminder

Addition rule: $\mathbb{P}(B \cup M) = \mathbb{P}(B) + \mathbb{P}(M) - \mathbb{P}(B \cap M)$ — subtract the overlap, which was counted twice. “Exactly one” = union – intersection.

Exercise 4 Choose 3 of 10 images (4 are faces).

Solution. There are $\binom{10}{3} = 120$ equally likely selections.

(a) All faces: $\binom{4}{3} = 4$ ways, so $\frac{4}{120} = \boxed{1/30 \approx 0.033}$.

(b) Exactly one face: choose 1 of 4 faces and 2 of the 6 non-faces: $\binom{4}{1}\binom{6}{2} = 4 \times 15 = 60$, so $\frac{60}{120} = \boxed{0.5}$.

Reminder

Drawing *without replacement* and counting favourable vs. total selections is the *hypergeometric* setup. $\binom{n}{k}$ counts unordered choices.

Exercise 5 The birthday paradox.

Solution.

(a) Line the n people up; each must avoid all previous birthdays:

$$\mathbb{P}(\text{no match}) = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{365 - n + 1}{365} = \frac{365!}{(365 - n)! 365^n}.$$

(b) For $n = 4$: $\mathbb{P}(\text{no match}) = \frac{365 \cdot 364 \cdot 363 \cdot 362}{365^4} = 0.9836$, so $\mathbb{P}(\text{match}) = 1 - 0.9836 = \boxed{0.0164}$ ($\approx 1.6\%$).

(c) For $n = 23$ the match probability is ≈ 0.507 . The reason it feels surprising: what matters is the number of *pairs*, and there are $\binom{23}{2} = 253$ of them, each with a $1/365$ chance to match.

Reminder

Our intuition counts *people*, but the probability is driven by *pairwise comparisons*, which grow like n^2 . This “comparisons, not items” idea recurs in combinatorics and in the complexity of algorithms.

Exercise 6 Signal detection: 100 old items (80 called “old”), 100 new (30 called “old”).

Solution.

(a) Hit rate $\mathbb{P}(\text{says old} \mid \text{old}) = \frac{80}{100} = \boxed{0.80}$.

(b) False-alarm rate $\mathbb{P}(\text{says old} \mid \text{new}) = \frac{30}{100} = \boxed{0.30}$.

(c) Of everyone who said “old” ($80 + 30 = 110$), 80 were genuinely old: $\mathbb{P}(\text{old} \mid \text{says old}) = \frac{80}{110} = \boxed{8/11 \approx 0.727}$.

Reminder

The hit rate $\mathbb{P}(\text{"old"} \mid \text{old})$ and the predictive value $\mathbb{P}(\text{old} \mid \text{"old"})$ are *different* conditionals: in general $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$. Swapping them is the “prosecutor’s fallacy”.

Exercise 7 Neuron A fires 40%, B fires 30%.

Solution.

- (a) If independent: $\mathbb{P}(\text{both}) = 0.4 \times 0.3 = 0.12$; $\mathbb{P}(\text{at least one}) = 1 - 0.6 \times 0.7 = 1 - 0.42 = 0.58$.
- (b) If $\mathbb{P}(\text{both}) = 0.20$, that exceeds the independent value 0.12, so the neurons are **not independent** — they co-fire more than chance (a positive *noise correlation*).

Reminder

Independence means $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ *exactly*. Any gap between the observed joint probability and the product signals a dependence.

Exercise 8 50% *easy* (0.95), 30% *medium* (0.80), 20% *hard* (0.55).

Solution. By the law of total probability,

$$\mathbb{P}(\text{correct}) = 0.5(0.95) + 0.3(0.80) + 0.2(0.55) = 0.475 + 0.24 + 0.11 = 0.825.$$

Reminder

The law of total probability is a *weighted average* of the conditional probabilities, each weighted by how often its scenario occurs.

Exercise 9 Prevalence 2%, sensitivity 95%, specificity 90%; test is positive.

Solution. Let $D =$ has the condition. Then

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\mathbb{P}(D)}{\mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid D^c)\mathbb{P}(D^c)} = \frac{0.95(0.02)}{0.95(0.02) + 0.10(0.98)} = \frac{0.019}{0.117} = \approx 0.162.$$

Because the condition is rare, the $0.10 \times 0.98 = 0.098$ false positives outnumber the 0.019 true positives.

Reminder

Natural-frequency check: out of 1000 people, 20 are ill (≈ 19 test positive) and 980 healthy (≈ 98 test positive), giving $19/117 \approx 16\%$. The *base rate* dominates the answer.

Exercise 10 Monty Hall.

Solution.

- (a) Staying wins only if your first pick was already correct: $\mathbb{P}(\text{win} \mid \text{stay}) = \boxed{1/3}$.
- (b) Switching wins whenever your first pick was wrong: $\mathbb{P}(\text{win} \mid \text{switch}) = \boxed{2/3}$.
- (c) The host *always* opens a losing door and is free to choose which; that constrained choice concentrates the remaining $2/3$ of the probability onto the door you can switch to.

Reminder

The paradox dissolves once you notice the host's action is *not* random: it depends on where the prize is, so it carries information you must condition on.

Exercise 11 75% accuracy, $n = 12$ independent trials; $X = \text{number correct}$.

Solution.

- (a) $X \sim \boxed{\text{Binomial}(12, 0.75)}$.
- (b) $\mathbb{E}[X] = np = 12(0.75) = \boxed{9}$; $\text{Var}(X) = np(1-p) = 12(0.75)(0.25) = \boxed{2.25}$ (SD = 1.5).
- (c) $\mathbb{P}(X = 12) = 0.75^{12} = \boxed{\approx 0.0317}$.

Reminder

A Binomial is a sum of n independent Bernoulli trials; mean np , variance $np(1-p)$. “All n correct” has probability p^n .

Exercise 12 Poisson spikes, $\lambda = 4$ per window.

Solution.

- (a) $\mathbb{P}(X = 2) = \frac{\lambda^2 e^{-\lambda}}{2!} = \frac{16e^{-4}}{2} = 8e^{-4} = \boxed{\approx 0.147}$.
- (b) $\mathbb{P}(X = 0) = e^{-4} = \boxed{\approx 0.0183}$.
- (c) $\text{Var}(X) = \lambda = \boxed{4}$. Comparing the empirical variance to the mean tests the Poisson assumption (over- or under-dispersion).

Reminder

Poisson: $\mathbb{P}(X = k) = \lambda^k e^{-\lambda} / k!$, with $\mathbb{E}[X] = \text{Var}(X) = \lambda$. The ratio variance/mean (the *Fano factor*) equals 1 for a Poisson process.

Exercise 13 Reaction times Normal, mean 600, SD 80.

Solution.

(a) $z = \frac{760 - 600}{80} = \boxed{2.0}$.

(b) Slower than 760 ms is the area above $+2\sigma$. Since 95% lies within $\pm 2\sigma$, the two tails share 5%, so the upper tail is $\boxed{\approx 2.5\%}$.

(c) 520 to 680 ms is 600 ± 80 , i.e. within $\pm 1\sigma$: $\boxed{\approx 68\%}$.

Reminder

Standardise with $z = (x - \mu)/\sigma$, then use the 68–95–99.7 rule. A value 2 SD from the mean is already fairly extreme.

Exercise 14 Prior odds 1 : 4; the detector fires (likelihood ratio 6).

Solution.

$$\text{posterior odds} = \text{likelihood ratio} \times \text{prior odds} = 6 \times \frac{1}{4} = \frac{6}{4} = \boxed{3 : 2}.$$

Converting to a probability: $\frac{3}{3+2} = \boxed{0.6}$.

Reminder

The odds form of Bayes: *posterior odds* = *prior odds* $\times$ *likelihood ratio*. Independent pieces of evidence multiply their likelihood ratios together.

Exercise 15 The two-child problem.

Solution. The equally likely families are $\{GG, GB, BG, BB\}$ (elder listed first).

(a) “At least one girl” rules out BB , leaving $\{GG, GB, BG\}$; only GG has two girls, so $\mathbb{P} = \boxed{1/3}$.

(b) “The elder is a girl” leaves $\{GG, GB\}$; only GG works, so $\mathbb{P} = \boxed{1/2}$.

(c) The two statements define *different conditioning sets* (three families vs. two), so they give different answers even though both mention “a girl”.

Reminder

Whenever you condition, write down *exactly* which outcomes remain. Vague wording (“a girl”) hides which set you mean — a frequent source of paradoxes.